ME23312 ENGINEERING THERMODYNAMICS
UNIT I – Basics, Zeroth and First Law • Lecture Notes

SESSION 4 • TUTORIAL 1

Tutorial 1: Problems on Displacement Work & the p–V Diagram

Six fully solved problems with method notes + practice set with answers

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Purpose of this tutorial. To build fluency in displacement-work calculations: choosing the correct formula from the process law, handling units, finding unknown end states from $pV^n = C$, treating multi-step paths, and recognising the free-expansion trap. Six fully solved problems and a practice set with answers are provided.

1. Formula Recall

Process	Law	Work $W_{1\text{-}2}$
Isochoric	V = C	0
Isobaric	p = C	$p(V_2 - V_1)$
Isothermal (ideal gas)	pV = C	$p_1V_1 \ln(V_2/V_1) = mRT \ln(V_2/V_1) = mRT \ln(p_1/p_2)$
Polytropic	$pV^n = C$	$(p_1V_1 - p_2V_2)/(n - 1)$
Adiabatic (reversible)	$pV^\gamma = C$	$(p_1V_1 - p_2V_2)/(\gamma - 1)$

KEY POINT Solution routine: sketch the p–V diagram → identify the process law → convert data to SI (bar → kPa ×100) → find any unknown end state → apply the formula → box the answer with sign and unit.

2. Worked Problems

WORKED EXAMPLE T1.1 — Constant-pressure expansion

A gas expands at a constant pressure of 3 bar from 0.05 m^3 to 0.20 m^3. Determine the work done by the gas.

Solution.

1. Process: isobaric ⇒ $W = p(V_2 - V_1)$.
2. p = 3 bar = 300 kPa; ΔV = 0.20 − 0.05 = 0.15 m^3.
3. W = 300 × 0.15 = 45 kJ; positive ⇒ done by the gas.

ANSWER W = +45 kJ

Note: kPa × m^3 = kJ directly — the cleanest unit set.

WORKED EXAMPLE T1.2 — Polytropic compression

Air, 0.3 m^3 at 1 bar, is compressed according to $pV^{1.3} = C$ to a final volume of 0.05 m^3. Find the final pressure and the work done.

Solution.

1. $p_2 = p_1(V_1/V_2)^n = 1 \times (0.3/0.05)^{1.3} = 6^{1.3} = 10.27$ bar = 1027 kPa.
2. $W = (p_1V_1 - p_2V_2)/(n-1) = (100\times0.3 - 1027\times0.05)/0.3$.
3. W = (30 − 51.35)/0.3 = −71.2 kJ; negative ⇒ work done on the air.

ANSWER $p_2 \approx 10.27$ bar | $W \approx -71.2$ kJ

Note: $6^{1.3} = e^{1.3 \ln 6} = e^{2.329} = 10.27$.

WORKED EXAMPLE T1.3 — Isothermal expansion (mass form)

0.5 kg of air (R = 0.287 kJ/kg·K) at 400 K expands isothermally until its volume doubles. Calculate the work done.

Solution.

1. $W = mRT \ln(V_2/V_1) = 0.5 \times 0.287 \times 400 \times \ln 2$.
2. $mRT = 57.4$ kJ; $\ln 2 = 0.6931$.
3. $W = 57.4 \times 0.6931 = 39.79$ kJ.

ANSWER **$W \approx +39.8$ kJ**

Note: For an ideal gas at constant T, $\Delta U = 0$ and hence $Q = W$ (first-law preview).

WORKED EXAMPLE T1.4 — Two-step process

A gas at 1 bar, 0.10 m^3 is (i) heated at constant pressure to 0.30 m^3, then (ii) heated at constant volume until the pressure reaches 3 bar. Find the total work done and sketch the p–V diagram.

Solution.

1. Step (i): $W_1 = p\Delta V = 100 \times 0.20 = 20$ kJ (horizontal line on p–V).
2. Step (ii): $W_2 = 0$ (vertical line; $dV = 0$).
3. $W_{total} = 20 + 0 = 20$ kJ.

ANSWER **$W_{total} = +20$ kJ**

WORKED EXAMPLE T1.5 — Work from a linear p-V relation

During an expansion the pressure varies linearly with volume from 500 kPa at 0.02 m^3 to 100 kPa at 0.10 m^3 (a spring-loaded piston). Find the work done.

Solution.

1. For a straight-line path, W = area of the trapezium under the line = mean pressure × ΔV.
2. $p_{mean} = (500 + 100)/2 = 300$ kPa; $\Delta V = 0.08$ m^3.
3. $W = 300 \times 0.08 = 24$ kJ.

ANSWER **$W = +24$ kJ**

Note: A linear p(V) arises whenever a linear spring backs the piston.

WORKED EXAMPLE T1.6 — Free expansion (concept trap)

A rigid, insulated vessel is divided by a partition: 0.05 m^3 of gas at 5 bar on one side, vacuum on the other. The partition is removed. How much displacement work is done by the gas?

Solution.

1. The gas meets no resisting pressure (vacuum): there is nothing to push against.
2. By definition of work, no weight could be raised in the surroundings; the boundary of the rigid vessel does not move.
3. $W = 0$, even though the gas volume doubles. The process is not quasi-static, so $\int p\, dV$ is inapplicable.

ANSWER **$W = 0$**

Note: The most common conceptual error in CAT answers — free expansion does no work.

3. Practice Problems (answers given)

#	Problem	Answer
P1	A piston–cylinder holds 0.1 m^3 of gas at 2 bar. Heat is added at constant pressure until V = 0.25 m^3. Find W.	+30 kJ
P2	A gas at 8 bar, 0.02 m^3 expands to 1 bar with $pV^{1.4} = C$. Find V_2 and W.	$V_2 \approx 0.0885$ m^3; $W \approx +17.9$ kJ
P3	1 kg of air at 320 K is compressed isothermally to one-third of its volume. Find W.	−100.9 kJ
P4	Pressure rises linearly from 100 kPa at 0.5 m^3 to 400 kPa at 0.2 m^3 during compression. Find W.	−75 kJ

4. Takeaways

- The process law selects the formula; never mix formulas across processes.
- Find unknown end states from the process law before computing work.
- Expansion ⇒ W positive; compression ⇒ W negative — let the algebra produce the sign.
- Linear p–V paths: use mean pressure × ΔV (trapezium area).
- Free expansion does zero work regardless of the volume change.

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 3 (Work and Heat Transfer). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.